

Sheet 5: Matrix Factorization

Exercise 1 (Matrix Factorization)

[10 Points]

We will continue to use the MovieLens dataset and LensKit and move from similarity-based methods to latent factor models.

- a) **3 Points** In a first step, train a matrix factorization model with LensKit, using LensKit's bi-ased matrix factorization model for explicit rating prediction. Explain the prediction model in terms of global average, user bias, item bias, and latent user/item factors. Then identify the most important hyperparameters, e.g., number of latent dimensions, number of iterations, regularization, and bias-related parameters.
- b) **2 Points** In a next step, we will focus on hyperparameters. Vary at least two important hyperparameters, such as the number of latent dimensions and regularization strength. Report how the changes affect prediction quality. Subsequently, discuss the results (e.g., do larger models always perform better? are there indications for under- or overfitting in your results?)
- c) **2 Points** Evaluate the matrix factorization model and compare it to the previous models from the last sheet. Perform the following steps:
 - Use k -fold cross-validation or a train/test split.
 - Compute rating prediction metrics such as RMSE and MAE, top- k recommendation metrics such as Precision, Recall, F_1 -score, or nDCG.
 - Clearly explain how you generate top- k recommendations from predicted ratings.
 - Discuss differences between rating-prediction quality and ranking quality.
- d) **3 Points** Visualize the learned user and item embeddings.
 - Extract the learned user and item latent factors.
 - Use dimensionality reduction, e.g., PCA, t-SNE, or UMAP, to reduce the embeddings to 2D.
 - Pick one user and visualize the user's position together with selected movies, including highly rated movies and recommended movies.
 - Discuss whether the 2D visualization meaningfully reflects recommendation candidates.
 - Elaborate on how much information is lost through dimensionality reduction and whether the plot should be interpreted cautiously.

Important: Submit your solution to OLAT and mark your solved exercises with the provided checkboxes. The deadline ends at 23:59 on the day before the discussion.